

**Research Question: Can we Predict the Direction of Future
Stock Price Movements in Year $t+1$ Using Firm Financials
and Economic Sentiment in Year t ?**

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Predicting Future Stock Price Movements Using Firm Financials and Economic Sentiment

1. Introduction

Financial markets continuously incorporate a mixture of both firm-specific fundamentals and macroeconomic expectations. This project investigates whether cross-sectional differences in firms' accounting characteristics (drawn from their respective U.S. Securities and Exchange Commission (SEC) tax filings) in year t , combined with broader economic sentiment indicators, can be used to predict whether a company's stock price will rise or fall in the following year, $t+1$. The ability to forecast price direction can yield important insights about market efficiency and the informational value of accounting data.

To study this, I build a firm-year panel for companies listed in the United States between fiscal years 2017 and 2019. The panel combines SEC EDGAR fundamentals, stock prices from Yahoo Finance, and the University of Michigan consumer sentiment index, linked through a SEC Central Index Key-ticker mapping. The outcome variable is a simple indicator for whether the firm's stock price at the end of year $t+1$ is higher than at the end of year t .

The project emphasises not only the prediction exercise itself, but also the data engineering needed to progress from the raw tax filings to a clean modelling dataset. In particular, I implement an imputation strategy that uses asset-size buckets within each year to fill missing accounting values, log transform any skewed variables, and construct economically meaningful ratios (leverage, profitability, liquidity). In terms of my prediction methods, I adopt a Random Forest classifier to model the probability of a price increase.

2. Data

2.1 Data Selection

This analysis uses 4 primary data sources.

1. SEC EDGAR "Financial Statements Data" (FSD)

I use the official SEC FSD zip archives for 2017Q4, 2018Q4, and 2019Q4, each of which contains sub.txt (submission metadata) and num.txt (numeric values). These files provide the universe of tagged accounting data for U.S. public companies. Raw accounting data that are obtained from the sub.txt and num.txt files in the EDGAR dataset. These include identifiers (CIK, fiscal years (fy)), and quantitative tags such as revenues, total assets, liabilities, cash flows, and net income.

2. Macroeconomic Sentiment Data

I load a CSV containing monthly values of the University of Michigan Consumer Sentiment Index (UMCSENT), navigate the date column, and compute yearly averages for 2017-2019. I then standardise this sentiment series (subtracting its mean and dividing by its standard deviation) to obtain an econ_sentiment z-score for each year.

3. SEC CIK-Ticker Mapping

A JSON file with company metadata is loaded into a DataFrame, from which I extract a unique mapping from CIK (our firm identifier) to stock ticker (the firm's identifier on the stock market). I then compare this list of CIK-Tickers to the CIKs in my model DataFrame and keep only those tickers that have a corresponding CIK identifier in my model DataFrame.

4. Yahoo Finance Stock Market Prices

The resulting tickers are used to download the daily adjusted prices from Yahoo Finance for 2017-2021 via the yfinance package. This provides me with my target variable.

2.2 From Tax Filings to Firm-Year Panel

The function 'build_features_for_period(period_code)' performs the core data construction for each quarter archive (2017Q4, 2018Q4, 2019Q4).

1. Filter to 10-K filings

From sub.txt, I identify the form, adsh, cik, and fy (fiscal year) columns and keep only 10-K-type filings (10-K refers to the tax filings for U.S companies). This yields a table of firm-year identifiers with (adsh, cik, fy) as keys.

2. Select accounting tags

In num.txt, I select a curated list of important accounting tags such as Revenues, Assets, Liabilities and several others, restricting to USD units and annual values (qtrs equal to 0 or 4)

3. Pivot to wide firm-year format

I merge num_filt with the 10-K subset on adsh, and then move into a wide table with index (cik, fy) and one column per tag. This produces an initial firm-year panel with raw level variables.

4. Restrict to relevant fiscal years

I keep only fiscal years 2017-2019, ensuring each observation corresponds to a firm-year with fundamentals from that same period.

2.3 Cleaning and Imputation with Size Buckets

A key feature of the process is the cleaning and imputation strategy that exploits firm size and time. This method is more robust than a simple median/mean value imputation for each tag, as it takes into account firm size (based on assets) before assigning a value.

- First, for each period I compute the fraction of missing values for each raw accounting tag and drop those with more than 25% NaNs (missing values), ensuring that only reasonably well-reported tags are retained.
- I then construct size buckets based on total assets. If the assets column is available, firms are split into quartiles of assets using pd.qcut, creating a size bucket within each period. If the assets column is missing for any reason, a dummy bucket is used to ensure that the code still runs.

For each remaining raw column, missing values are filled as follows:

1. Size bucket x year median

Within each bucket and fiscal year, I compute the median of the variable and use it to fill missing entries. This means a firm is primarily imputed using other firms of similar size in the same year.

2. Year-only median

If the size bucket median is missing, I fall back to the overall median within the same fiscal year.

3. Global median

As a final fallback, and to ensure that the programme runs, I replace any remaining missing values with the overall median of that variable in the overall period.

I carry this procedure out twice. First to the raw variables (referred to as tags above), and later to the ratio variables (I will discuss these further below). In both cases, the temporary size bucket column is dropped after imputation so it does not enter the model directly.

2.4 Transformations and Ratio Construction

After imputing values into the raw tags, I examine their skewness and identify those with a strong right skew. For those variables, I create log-transformed versions using `np.log1p` on the non-negative values, resulting in new columns named `log_assets`, `log_revenues`, `log_liabilities` etc. Note that the code is unable to transform negative values, a drawback that is discussed in the limitations section below.

I then generate a set of financial ratios including but not limited to the following:

- Leverage and Capital Structure: $\text{leverage} = \text{liabilities} / \text{assets}$
- Liquidity: $\text{current_ratio} = \text{assets current} / \text{liabilities current}$

These derived ratios are again subjected to the ‘size bucket $\times$ year’ imputation described in section 2.3. Finally, at the modelling stage I ‘winnerise’ key ratio variables at the 1st and 95th percentiles to reduce the influence of extreme outliers.

2.5 Linking to Prices and Sentiment

Using the SEC CIK-ticker mapping dataset, I form a unique integer CIK (and it’s relative ticker) to the CIKs in my DataFrame. I then download daily adjusted prices for all tickers in my period, compute year-end closing prices, and build a prices panel containing `price_fy` (year t) and `price_next` (year $t+1$) for fiscal years 2017-2019 (dropping any tickers for which prices are failed to be pulled).

Merging this with the SEC financial panel yields the outcome we get:

$$\text{price_up_next} = 1 \text{ (price_next} > \text{price_fy)}$$

Finally, I merge the yearly standardised sentiment (`econ_sentiment`) onto the firm-year panel by fiscal year and use the resulting `econ_sentiment` as a macro feature.

3. Methods

3.1 Feature Set and Target

For modelling, I construct `df_model` from the full panel, which now includes both prices and economic sentiment. The binary target variable is:

- `Price_up_next` = 1 if next-year year-end price exceeds current year-end price, 0 otherwise.

I focus on a curated set of numeric predictors (the tags, ratios and sentiment outlined above). I exclude identifiers (`cik_int`, `fy`), price variables, and the target from the feature set (to avoid look ahead bias), and drop any features that are entirely missing or constant (no variation would be irrelevant for the prediction). Before fitting the model, any remaining missing values in the chosen features are filled using simple median imputation). This provides us with a compact yet economically interpretable predictor set.

3.2 Train and Test Split and Baseline

The `'train_test_split'` is then used to split the cleaned data into training (70%) and test (30%) sets, stratifying by price direction in year $t+1$ to preserve the share of up/down outcomes. A simple baseline classifier always predicts the majority class in the training set on the test set. I compute the majority-class share in the training data and its corresponding accuracy on the test data: this is the “no skill” benchmark.

3.3 Random Forest Classifier

The predictive model is a Random Forest classifier from `sklearn.ensemble`. I provide the model with the following specifications:

- `n_estimators` = 500. Sets the number of trees in the forest. More trees results in a longer training time.
- `Max_depth` = None. Allows the model to capture complex patterns.
- `Random_state` = 42. For reproducibility. Allows for comparable results.

The model is trained on the training set and evaluated on the test set.

3.4 Rationale

The Random Forest is well-suited to this problem for several reasons:

- It is robust to moderate multicollinearity.
- Feature importance measures provide insight into which fundamentals and macro variables are most influential for predicting stock price direction in year $t+1$. An important aspect for this economic question.

- The baseline also provides a transparent benchmark. If the Random Forest cannot beat this simple strategy, then the data and methodology are not extracting meaningful predictive signals.

4. Results and Interpretation

4.1 Model Performance vs Baseline

The majority-class baseline achieves an accuracy of 0.545, which is equal to the share of the dominant outcome in the training set . On the test set, this translates into a baseline accuracy that essentially reflects how often firms' prices go up (or down) in the chosen period.

The Random Forest clearly outperforms this baseline on the held-out test set, with an accuracy of 0.64. The reported Random Forest accuracy is higher than the majority-class accuracy, demonstrating that the combined fundamentals and sentiment features contain predictive information about price direction in year $t+1$. However, the predictive power is modest, consistent with the idea that equity markets are largely, though not perfectly, efficient.

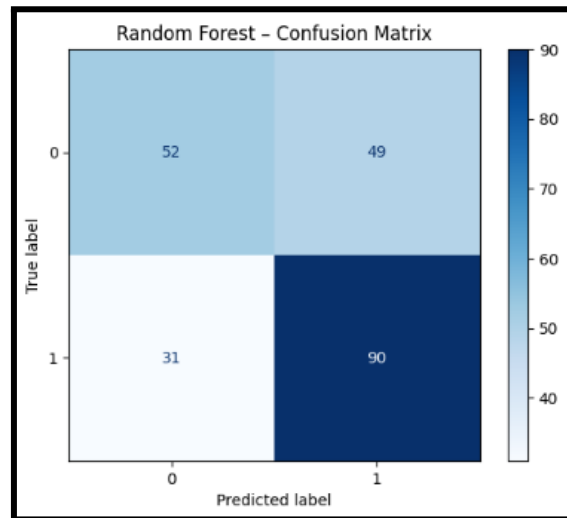
4.2 Other Measures of Model Strength

Table. 1
Classification Report

	Precision	Recall	F1-Score	Support
0	0.63	0.51	0.57	101
1	0.65	0.74	0.69	121
Accuracy			0.64	222
Macro avg	0.64	0.63	0.63	222
Weighted avg	0.64	0.64	0.63	222

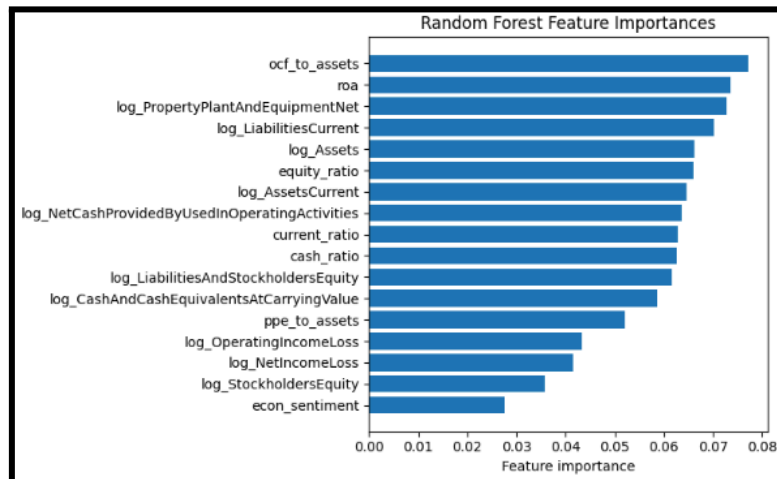
The classification report above gives us a summary as to how well the Random Forest predicts whether a firm's stock price will rise in year $t+1$. These results suggest that the model is more effective at identifying case 1s than 0s, while still performing better than a naive baseline model, which has an accuracy of 0.545.

Figure. 1
Confusion Matrix



This stronger case for class 1 prediction is reflected in the confusion matrix, which details that the model correctly predicts 52 price decreases and 90 price increases, while misclassifying 49 price decreases as increases, and 31 increases as decreases. While these errors aren't wildly lopsided, the larger number of false positives suggests that there is a tendency to label borderline cases as price increases.

Figure. 2
Feature importance



The feature importance bar chart provides us with a ranking of the variables that contribute to predicting stock direction in year $t+1$, arranged from most to least important. Size and asset measures appear near the top, while leverage variables and macroeconomic sentiment play relatively non-trivial roles in prediction. This suggests that the model relies most heavily on firm size and balance sheet strength indicators. However, this does not mean that these features are model specific, and should not be thought of as causal effects.

5. Discussion

5.1 Policy Implications

Even though the predictive power of the model is modest, the findings have several policy implications. First, the fact that publicly available fundamentals (and macro sentiment to some extent) can systematically predict next-year price direction suggests that raw information is not always instantly and perfectly incorporated into prices (financials take time to reach the market). Regulators could therefore view improvements in disclosure quality and accessibility as a way to enhance market efficiency. Clearer, more standardised financial reporting may help less sophisticated investors process information that professional models already exploit.

Since these relatively standard financial variables and ratios can be combined into models with some level of stock price prediction, policymakers may need to consider governing the use of ML models in retail or professional investing. For example, retail investing platforms with 'AI investment recommendations' might be required to disclose the models they are running, which would help to prevent biased or overconfident claims of predictive power.

5.2 Limitations

Several limitations should be acknowledged:

- The sample covers only three fiscal years and a period of relatively stable macroeconomic conditions. The relationships may differ in crisis periods (2007-2010 for example).
- The target is binary, so the model does not distinguish between small and large price moves.
- The inability to transform negative values reduces the models ability to compare across firms, or capture financially distressed firms. This also results in overrepresentation of 0 in value distributions.
- Despite imputation, missingness may still be systematically related to firm health, which could bias both the predictive model and economic interpretation.

5.3 Extensions

Building on those limitations, future work could extend this framework in several directions:

- Expand the time window beyond 2017-2019. Test whether predictive performance holds in stress periods.
- Predict magnitude and not just direction of prices.

- Add richer macroeconomic variables. Aspects like sector specific indices or inflation/interest rates could add to the predictive power as these are features that strongly influence investment decisions.

6. Conclusion

This project investigated whether company financials and macroeconomic sentiment can predict whether a firm's stock price will rise over the next year. Using SEC EDGAR financial statement data for 2017-2019, linked to stock prices via a CIK-ticker mapping and supplemented with UMICH consumer sentiment, I constructed a cleaned firm-year panel with log-transformed and winorised ratios, and a carefully imputed set of features based on asset-size buckets within each year.

A Random Forest classifier trained on these features outperforms a simple majority-class baseline on a held-out test set, indicating that fundamentals and sentiment together hold predictive power for one-year-ahead price direction. The most important variables include a firm size and balance sheet indicators. Beyond the specific prediction results, the project demonstrates how big data methods, combined with careful data cleaning and transformation, can be used to test classic questions in economics within the structure of an applied research report.

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